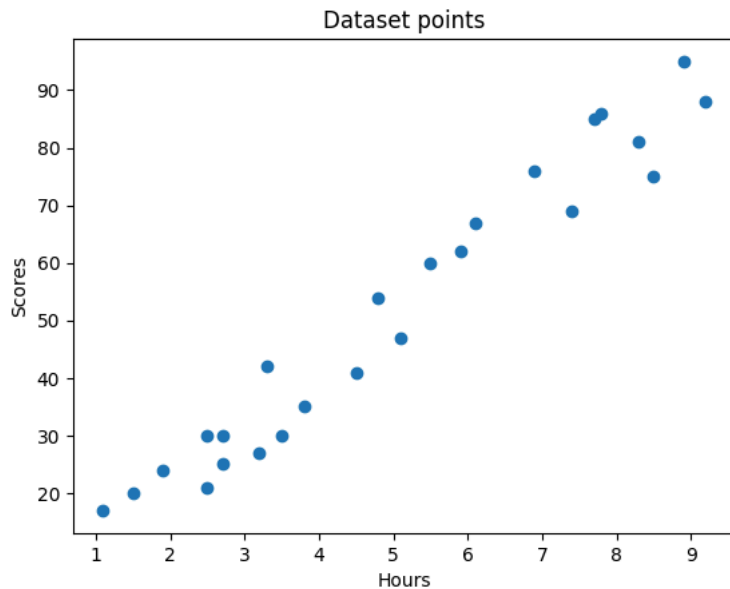


```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
dataset = pd.read_csv('Student_scores.csv')
plt.scatter(dataset['Hours'],dataset['Scores'])
plt.xlabel('Hours')
plt.ylabel('Scores')
plt.title('Dataset points')
plt.show()
```



```
X = dataset.iloc[:, :-1].values
Y = dataset.iloc[:, -1].values
```

```
print(Y.shape)
```

```
(25,)
```

```
print(X.shape)
```

```
(25, 1)
```

```
m = (90 - 20) / (9 - 2)
print(m)
c = 20 - m * 2
print(c)
print(f"y = {m}x + {c}")
```

```
10.0
0.0
y = 10.0x + 0.0
```

```
from sklearn.model_selection import train_test_split
X_train, X_test, Y_train, Y_test = train_test_split(X,Y,test_size=0.2,random_state=0)
```

```
from sklearn.preprocessing import PolynomialFeatures
poly = PolynomialFeatures(degree = 3)
X_poly = poly.fit_transform(X)
print(X_poly)
```

```
[[ 1.    2.5    6.25  15.625]
 [ 1.    5.1   26.01  132.651]
 [ 1.    3.2   10.24   32.768]
 [ 1.    8.5   72.25  614.125]
 [ 1.    3.5   12.25   42.875]
 [ 1.    1.5    2.25    3.375]
 [ 1.    9.2   84.64  778.688]
 [ 1.    5.5   30.25  166.375]
 [ 1.    8.3   68.89  571.787]
 [ 1.    2.7    7.29   19.683]
 [ 1.    7.7   59.29  456.533]
```

```
[ 1.    5.9   34.81 205.379]
[ 1.    4.5   20.25  91.125]
[ 1.    3.3   10.89  35.937]
[ 1.    1.1    1.21   1.331]
[ 1.    8.9   79.21 704.969]
[ 1.    2.5    6.25  15.625]
[ 1.    1.9    3.61   6.859]
[ 1.    6.1   37.21 226.981]
[ 1.    7.4   54.76 405.224]
[ 1.    2.7    7.29  19.683]
[ 1.    4.8   23.04 110.592]
[ 1.    3.8   14.44  54.872]
[ 1.    6.9   47.61 328.509]
[ 1.    7.8   60.84 474.552]]
```

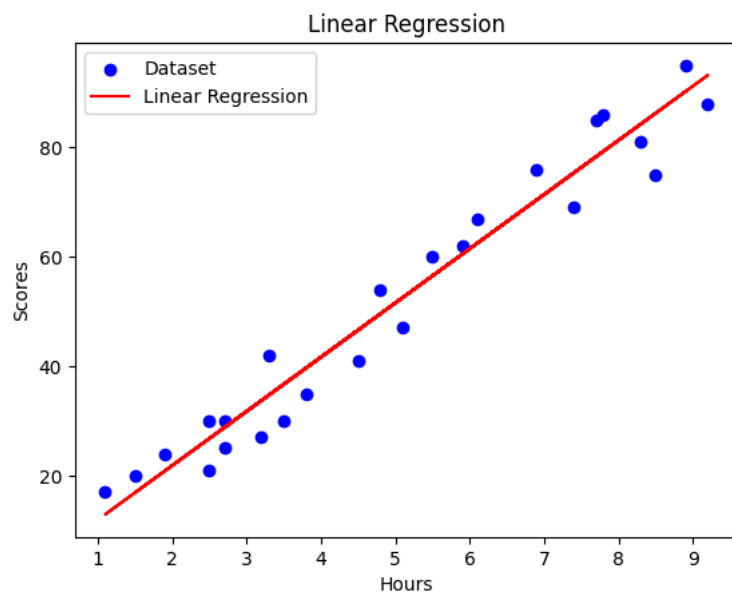
```
from sklearn.linear_model import LinearRegression
regressor = LinearRegression()
regressor.fit(X_train,Y_train)
m = regressor.coef_
c = regressor.intercept_
print(f"y = {m}x + {c}")
```

y = [9.91065648]x + 2.018160041434662

```
plt.scatter(X, Y, color='blue', label='Dataset')

plt.plot(X, regressor.predict(X), color='red', label='Linear Regression')

plt.xlabel("Hours")
plt.ylabel("Scores")
plt.title("Linear Regression")
plt.legend()
plt.show()
```



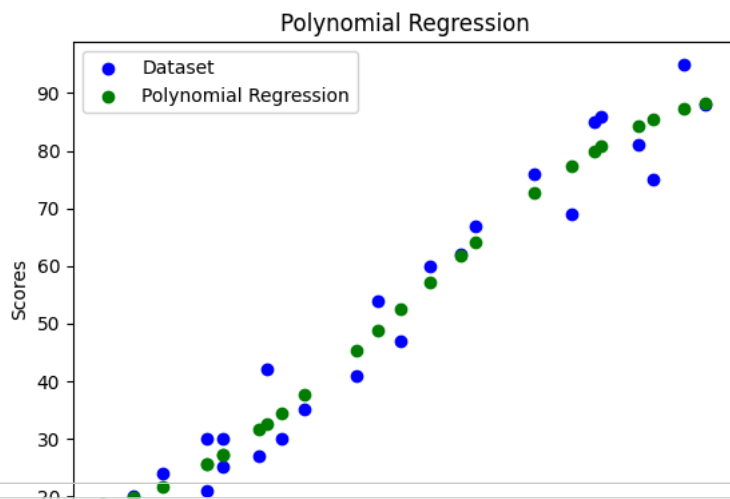
```
poly_regressor = LinearRegression()
poly_regressor.fit(X_poly,Y)
m1,m2,m3 = poly_regressor.coef_[0], poly_regressor.coef_[1], poly_regressor.coef_[2]
c = poly_regressor.intercept_
print(f"y = {m1}x^1 + {m2}x^2 + {m3}x^3 + {c}")
```

y = 0.0x^1 + -3.7970591287526343x^2 + 2.998747749044057x^3 + 19.317706990502344

```
plt.scatter(X, Y, color='blue', label='Dataset')

plt.scatter(X, poly_regressor.predict(X_poly), color='green', label='Polynomial Regression')

plt.xlabel("Hours")
plt.ylabel("Scores")
plt.title("Polynomial Regression")
plt.legend()
plt.show()
```



```
print('Final all 3 graphs')
plt.scatter(X, Y, color='blue', label='Dataset')

plt.plot(X, regressor.predict(X), color='red', label='Linear Regression')
plt.scatter(X, poly_regressor.predict(X_poly), color='green', label='Polynomial Regression')

plt.xlabel("Hours")
plt.ylabel("Scores")
plt.title("Comparison of all 3 graphs")
plt.legend()
plt.show()
```

Final all 3 graphs

